FACE IMPLEMENTATION

PAHILE BASIC TRANSFER LEARNING MODELS LA TRAIN KELA APAN NE

Model	**Accuracy**	**Training time**
ResNet50	~77%	1-2 hrs
VGG16	~76%	1-2 hrs
InceptionV3	~68%	1-2 hrs

NANTAR ACCURACY KAMI YET HOTI TAR APAN NE DATA PREPROCESSING AANI EXPLORATION KELA

```
disgust: train=349, val=87
angry: train=3196, val=799
fear: train=3278, val=819
neutral: train=3972, val=993
surprise: train=2537, val=634
happy: train=5772, val=1443
sad: train=3864, val=966
```

```
Total duplicate files found: 1236
Total unreadable files (skipped): 0
```

	Phase	Class	After (preprocessed_images)	Before (original_images)
0		angry	3059.0	3196.0
1		disgust	298.0	349.0
2		fear	3086.0	3278.0
3		happy	5648.0	5772.0
4		neutral	3881.0	3972.0
5		sad	3753.0	3864.0
6		surprise	2065.0	2537.0

```
check_image_size(folder_path)
```

```
Unique image size in /kaggle/working/original_images : []
Unique image size in /kaggle/working/original_images/validation : []
Unique image size in /kaggle/working/original_images/validation/disgust : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/angry : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/fear : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/neutral : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/surprise : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/happy : [(48, 48)]
Unique image size in /kaggle/working/original_images/validation/sad : [(48, 48)]
Unique image size in /kaggle/working/original_images/train : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/disgust : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/angry : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/fear : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/neutral : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/surprise : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/happy : [(48, 48)]
Unique image size in /kaggle/working/original_images/train/sad : [(48, 48)]
```

```
df_sorted
```

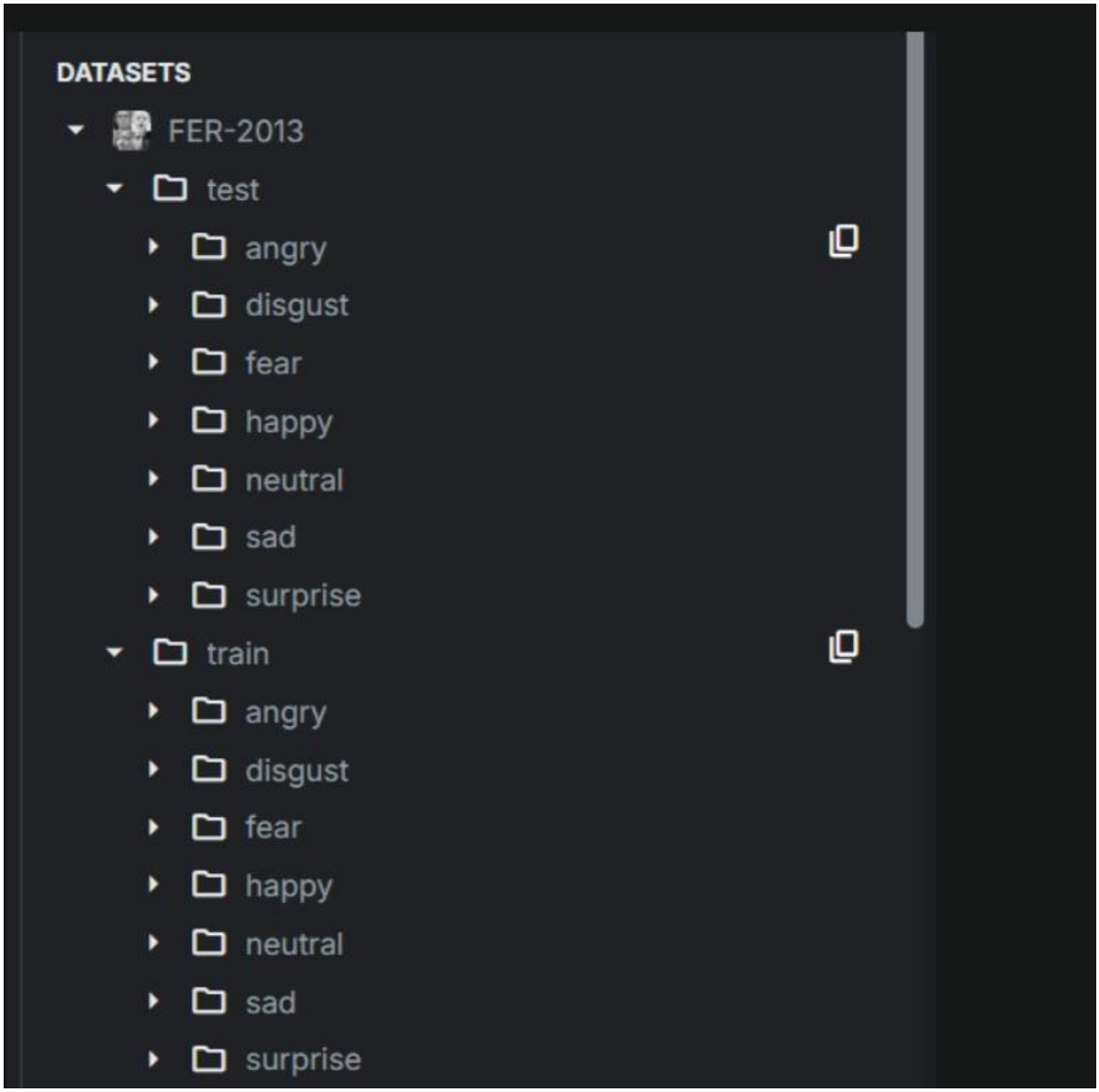
[14]:

	file_name	folder	size_kb
11267	Training_87394233.jpg	train/fear	2.34
9284	Training_98835403.jpg	train/angry	2.31
4199	Training_9066250.jpg	validation/happy	2.28
6971	Training_58690193.jpg	train/angry	2.26
16823	Training_82417057.jpg	train/surprise	2.25
...	...	...	...
8206	Training_78540321.jpg	train/angry	0.35
28303	Training_48621797.jpg	train/sad	0.35
17523	Training_48403842.jpg	train/surprise	0.35
20960	Training_87607167.jpg	train/happy	0.35
9226	Training_10131352.jpg	train/angry	0.35

28709 rows × 3 columns







NANTAR MAGH CLASS IMBALANCE SOLVE KARNA SAATHI SMOTE METHOD USE KELA

2000 PER EMOTION CLASS

https://www.kaggle.com/code/aakarsonawane/notebookb8b7cdb763

1. Implementation details

Idea: teen frozen ImageNet backbones se features nikalke concatenate karna, phir ek chhota MLP classifier train karna.

- **Feature extractors (frozen, pooling="avg"):** VGG16 (512-d), ResNet50 (2048-d), InceptionV3 (2048-d)
- **Fused vector:** 512 + 2048 + 2048 = **4608-d**

- **Classifier:** Dense(512, ReLU) → BatchNorm → Dropout(0.5) → Dense(256, ReLU) → BatchNorm → Dropout(0.4) → Dense(7, softmax)
- **Training:** Adam (lr 1e-3), sparse categorical crossentropy, batch 64, max 40 epochs
- **Callbacks:** ModelCheckpoint (best val_accuracy), EarlyStopping (val_loss, patience 8, restore best weights), ReduceLROnPlateau (factor 0.5, patience 3)
- **Environment:** Kaggle, Tesla T4 GPU, TensorFlow/Keras, imbalanced-learn
- Features .npy files mein save kiye taaki teammate dobara extraction na kare (3000/class run ke liye)

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2. Data preprocessing

1. Dataset: FER-2013 (msambare version, folder structure), 7 classes
2. Original **train folder** ko 80-20 split kiya (seed 42) → train aur validation. Official test folder use nahi hua.
3. **MD5 hashing se duplicate removal:** 1,236 duplicate images delete (28,709 → 27,473)
4. Corrupt images check (0), file format check (sab .jpg), size check (sab 48×48)
5. Grayscale → RGB, **bicubic resize 48×48 → 224×224**
6. Model-specific preprocess_input (VGG16, ResNet, Inception ke apne)
7. **Target-count resampling (sirf train par):** bade classes ko random undersample, chhote classes ko **SMOTE** (k=5) se 2000/class tak. Val set bilkul untouched.

Stage	Count
After dedup	27,473
Train / Val	22,147 / 5,326
After resampling	14,000 (2000 × 7)

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3. Exploration (EDA) methods

- Har class ki 5 random sample images (sab 7 classes)
- Class distribution bar charts (train aur val)

- Duplicate detection (MD5), corrupt-file check, format check, image-size check, file-size anomaly analysis (1.0-2.3 KB, koi anomaly nahi)
- Before/after dedup class count comparison table

Train class counts after dedup (correct names): Angry 3075, Disgust **318**, Fear 3141, Happy 5688, Neutral 3916, Sad 3798, Surprise 2211. Disgust bahut chhoti class hai, isliye uska SMOTE se ~6x oversample hua.

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4. Results

Final fusion classifier (2000/class SMOTE), real val set (5,326 images):

Metric	**Value**
Accuracy	**37.96%**
Macro F1	0.3131
Weighted F1	0.3714
Macro precision / recall	0.3286 / 0.3493

Per-class (naam fix karke, alphabetical order ke hisaab se):

Class	**Precision**	**Recall**	**F1**	**Support**
Angry	0.3249	0.2362	0.2736	762
Disgust	0.0229	0.0794	0.0356	63
Fear	0.2710	0.1515	0.1943	746
Happy	0.6109	0.4732	0.5333	1397
Neutral*	0.3496	0.5251	0.4197	956
Sad*	0.3984	0.2186	0.2824	933
Surprise*	0.3225	0.7612	0.4530	469

*Notebook ke printed report mein ye teen naam galat the; yahan support se match karke sahi kiye hain (956 = Neutral, 933 = Sad, 469 = Surprise).

Training history: 12 epochs chale, EarlyStopping ne roka. Best val_loss epoch 4 par thi, wahi weights restore hue (val acc 37.96%).

Epoch	Train acc / loss	Val acc / loss
1	0.304 / 2.109	0.304 / 1.883
4	0.510 / 1.293	**0.380 / 1.617**
8	0.608 / 1.047	0.397 / 1.622
10	0.625 / 0.988	0.409 / 1.620
12	0.665 / 0.902	0.411 / 1.634

Observations (honest)

1. **Fusion ka accuracy (37.96%) individual models (68-77%) se bahut kam hai.** Ye report mein chhupana nahi, seedhe likho aur reason batao.
2. **Bada overfitting gap:** train accuracy 66% tak gayi, val 41% par atki. 4608-d features par 14,000 samples ke saath MLP jaldi yaad kar leta hai.
3. **Frozen ImageNet features FER-2013 ke liye kamzor hain:** 48×48 grayscale faces ko 224×224 par upscale karke RGB banana domain mismatch hai. Fine-tuning ke bina facial-expression ke subtle features nahi pakde jaate. Aapke individual models (jo 67-77% par gaye) shayad fine-tuned the; ye confirm unke notebooks se hoga.
4. **Disgust class fail:** F1 0.036. Sirf 318 real images the, SMOTE ne 1,682 synthetic banaye jo high-dimensional space mein real distribution jaise nahi hote. Model ne synthetic patterns seekhe, real nahi.
5. **Happy sabse behtar** (F1 0.53), jo FER-2013 mein hamesha hota hai (sabse zyada data, sabse clear expression). Fear aur Angry kamzor.
6. **SMOTE ne recall aur precision ka trade-off bigada:** Surprise ka recall 0.76 hai par precision 0.32. Model Surprise bahut zyada predict kar raha hai, kyunki training mein har class 2000 thi par val mein Surprise sirf 469 (~9%) hai. Balanced training aur imbalanced val ka mismatch.
7. **Methodology:** val set se hi early stopping, checkpoint aur final result dono nikle, aur official test folder use nahi hua. Report mein final result ke liye official test folder par ek baar evaluate karna behtar hoga.

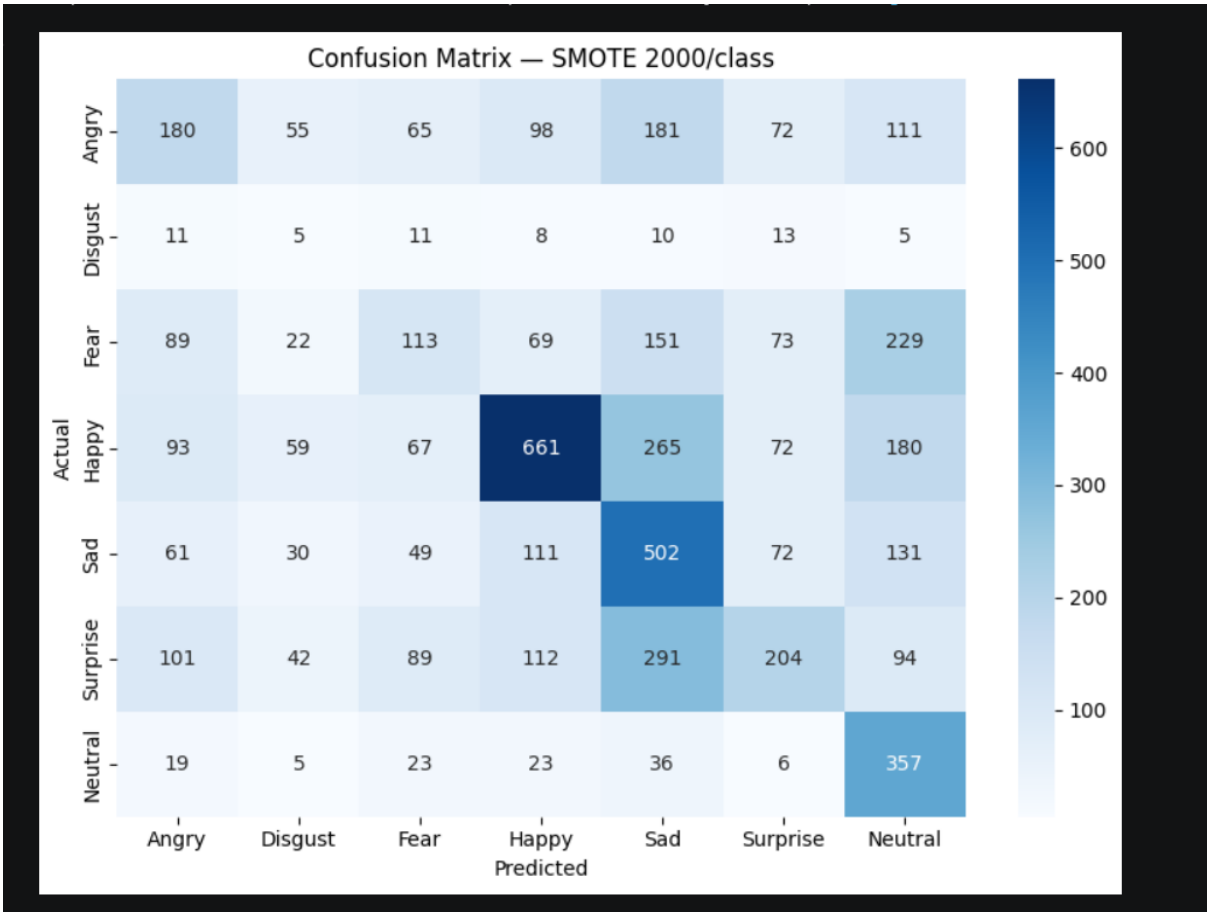
```
Epoch 1/40
219/219 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.2646 - loss: 2.3754
Epoch 1: val_accuracy improved from None to 0.30379, saving model to fusion_smote_2000perclass.keras

Epoch 1: finished saving model to fusion_smote_2000perclass.keras
219/219 ━━━━━━━━━━━━━━━━━━━━ 8s 20ms/step - accuracy: 0.3041 - loss: 2.1094 - val_accuracy: 0.3038 - val_loss: 1.8830 - learning_rate: 0.0010
Epoch 2/40
217/219 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.4113 - loss: 1.6152
Epoch 2: val_accuracy improved from 0.30379 to 0.34472, saving model to fusion_smote_2000perclass.keras

Epoch 2: finished saving model to fusion_smote_2000perclass.keras
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.4196 - loss: 1.5767 - val_accuracy: 0.3447 - val_loss: 1.7155 - learning_rate: 0.0010
Epoch 3/40
208/219 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.4710 - loss: 1.3994
Epoch 3: val_accuracy did not improve from 0.34472
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.4725 - loss: 1.4014 - val_accuracy: 0.3397 - val_loss: 1.7520 - learning_rate: 0.0010
Epoch 4/40
217/219 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.5168 - loss: 1.2893
Epoch 4: val_accuracy improved from 0.34472 to 0.37965, saving model to fusion_smote_2000perclass.keras

Epoch 4: finished saving model to fusion_smote_2000perclass.keras
219/219 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.5096 - loss: 1.2925 - val_accuracy: 0.3796 - val_loss: 1.6167 - learning_rate: 0.0010
Epoch 5/40
211/219 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.5397 - loss: 1.2250
Epoch 5: val_accuracy did not improve from 0.37965
...
    accuracy                           0.3796      5326
   macro avg       0.3286    0.3493    0.3131      5326
weighted avg       0.4059    0.3796    0.3714      5326

Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```



Conclusion (ready to use): *"Feature-level fusion of frozen VGG16, ResNet50 and InceptionV3 embeddings with SMOTE-based class balancing reached only 37.96% validation accuracy on FER-2013, well below the fine-tuned individual models (68-77%). Frozen ImageNet features do not transfer well to 48x48 grayscale facial expressions, and SMOTE on high-dimensional features could not rescue the minority Disgust class (F1 0.04)."*

Future scope: fine-tune backbones before fusion, class weights instead of SMOTE, weighted fusion of final predictions (late fusion), official test-set evaluation.

3000 PER EMOTION CLASS

https://www.kaggle.com/code/aakarsonawane/notebook3bca13f520

1. Implementation details

Wahi fusion pipeline: frozen VGG16 (512) + ResNet50 (2048) + InceptionV3 (2048) features → concat 4608-d → Dense(512) → BN → Dropout(0.5) → Dense(256) → BN → Dropout(0.4) → Dense(7, softmax).

Parameter	**Value**
Optimizer / Loss	Adam (1e-3) / sparse categorical crossentropy
Batch size / Max epochs	64 / 40
Callbacks	ModelCheckpoint (val_accuracy), EarlyStopping (val_loss, patience 8, restore best), ReduceLROnPlateau (×0.5, patience 3)
Target per class	**3000 (total 21,000)**
Resampling	RandomUnderSampler (bade classes) + SMOTE k=5 (chhote classes), sirf train par

2. Data preprocessing

Upstream wahi hai (dedup 1,236 images, 80-20 split, grayscale→RGB, 224×224 bicubic, model-specific preprocessing). Is notebook mein naya sirf resampling:

Class	**Before**	**After**
Angry	3075	3000
Disgust	318	3000 (SMOTE, ~9.4x)
Fear	3141	3000
Happy	5688	3000
Neutral*	3916	3000
Sad*	3798	3000
Surprise*	2211	3000

*Notebook ke printed naam galat hain, ye sahi mapping hai. Disgust ke 318 real samples se 2,682 synthetic banane padte hain, isliye ye aur zyada artificial hai.

3. EDA

Is notebook mein naya EDA nahi hai. Class distribution, duplicates, image sizes wagairah Aakar ke notebook mein hue the. Sirf before/after resampling counts print hote hain.

4. Results

Final (3000/class), real val set, 5,326 images: Accuracy **43.37%**, Macro F1 **0.3552**, Weighted F1 **0.4215**.

Class	Precision	Recall	F1	Support
Angry	0.3615	0.2415	0.2895	762
Disgust	0.0288	0.0476	0.0359	63
Fear	0.3821	0.1716	0.2368	746
Happy	0.5947	0.6271	**0.6105**	1397
Neutral*	0.3650	0.5460	0.4376	956
Sad*	0.3600	0.3430	0.3513	933
Surprise*	0.4727	0.5906	0.5251	469

Training history: 13 epochs chale, best val_loss epoch 5 (1.4934) par thi, wahi weights restore hue.

Epoch	Train acc / loss	Val acc / loss	LR
1	0.334 / 1.948	0.341 / 1.677	1e-3
3	0.498 / 1.318	0.411 / 1.561	1e-3
5	0.537 / 1.203	**0.434 / 1.493**	1e-3
9	0.603 / 1.043	0.434 / 1.521	5e-4
13	0.652 / 0.915	0.432 / 1.599	2.5e-4

2000/class vs 3000/class:

Metric	2000/class	3000/class	Change
Accuracy	37.96%	**43.37%**	+5.41
Macro F1	0.3131	**0.3552**	+0.04
Weighted F1	0.3714	0.4215	+0.05
Disgust F1	0.0356	0.0359	~0
Surprise* F1	0.4530	0.5251	+0.07
Happy F1	0.5333	0.6105	+0.08

Observations

1. **Zyada samples/class se accuracy 5.4% badhi.** 3000 mein kam undersampling hoti hai (Happy 5688→3000 vs 2000), toh zyada real data bacha. Ye ek achha point hai: sample count ka asar dikhta hai.
2. **Phir bhi 43% individual models (68-77%) se bahut kam hai.** Wajah wahi hai: frozen ImageNet features FER-2013 ke 48×48 grayscale faces ke liye weak hain. Fusion se ye gap nahi bhara.
3. **Disgust ab bhi fail** (F1 0.036), SMOTE ka count 2000 se 3000 karne se koi fayda nahi hua. 318 real images se synthetic samples real distribution ko represent nahi karte.
4. **Overfitting jaari:** train acc 65% tak, val 43% par flat. Epoch 5 ke baad val loss badhne lagta hai.
5. **Class imbalance ka asar kam hua:** Surprise ka precision-recall balance (0.47 / 0.59) ab 2000/class wale (0.32 / 0.76) se behtar hai.
6. **Val set se hi checkpoint aur result,** official test folder par evaluation nahi hua. Report mein ye limitation likho.

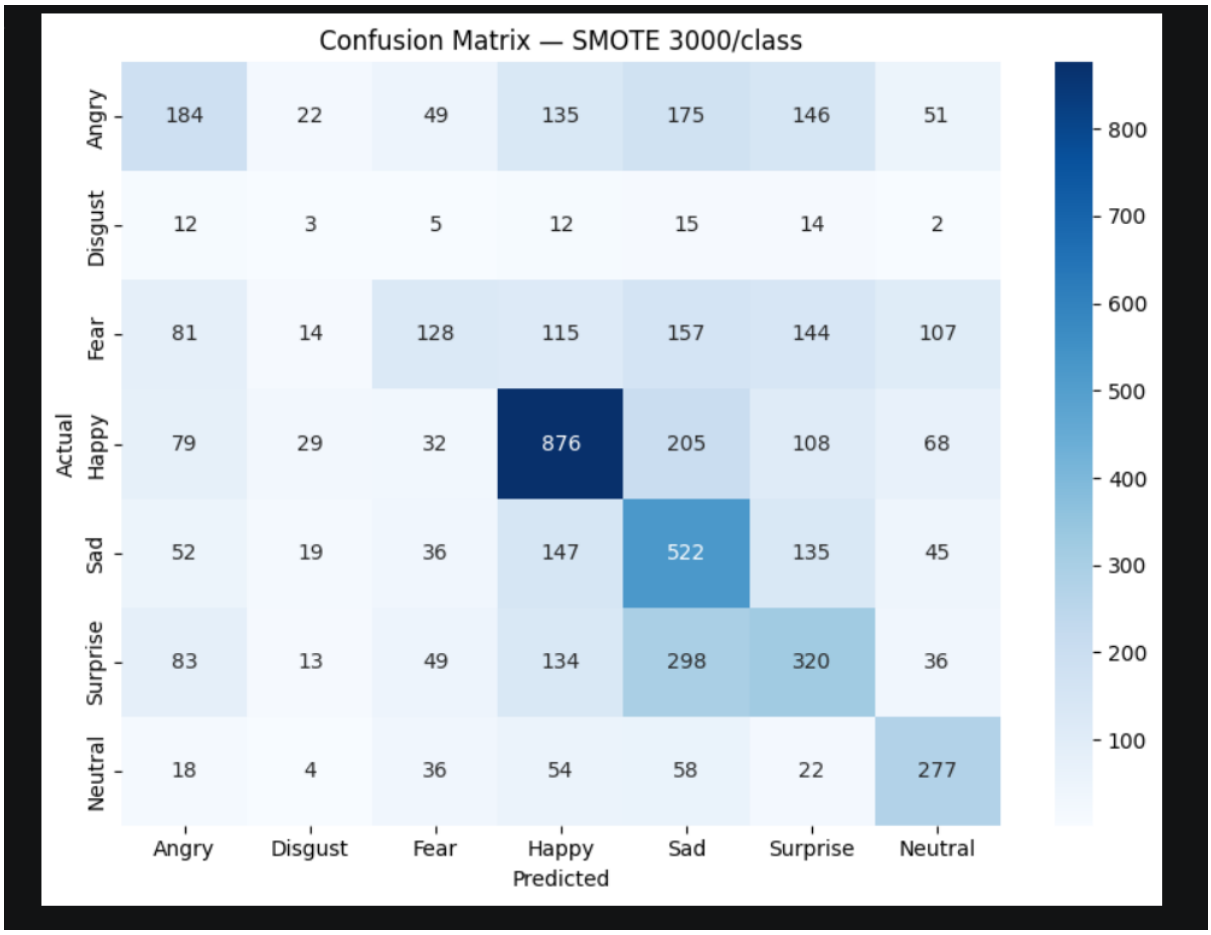
Conclusion (ready to use): *"Increasing the SMOTE target from 2000 to 3000 samples per class improved fusion accuracy from 37.96% to 43.37% (macro F1 0.313 to 0.355), but remained far below the individual fine-tuned models (68-77%). The minority Disgust class stayed unlearnable (F1 ~0.04), indicating that synthetic oversampling of high-dimensional frozen features cannot substitute for real data."*

```
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1788544180.389072      23 gpu_device.cc:2020] Created device /job:localhost/replica:0/task:0/device:GPU:0 with 13756 MB memory:  -> device: 0, name: Tesla T4, pci bus id:
I0000 00:00:1788544180.392492      23 gpu_device.cc:2020] Created device /job:localhost/replica:0/task:0/device:GPU:1 with 13756 MB memory:  -> device: 1, name: Tesla T4, pci bus id:
Epoch 1/40
 45/329 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.2217 - loss: 2.6121
I0000 00:00:1788544187.762007      73 device_compiler.h:196] Compiled cluster using XLA!  This line is logged at most once for the lifetime of the process.
329/329 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.2891 - loss: 2.2042
Epoch 1: val_accuracy improved from None to 0.34059, saving model to fusion_smote_3000perclass.keras

Epoch 1: finished saving model to fusion_smote_3000perclass.keras
329/329 ━━━━━━━━━━━━━━━━━━━━ 10s 15ms/step - accuracy: 0.3338 - loss: 1.9477 - val_accuracy: 0.3406 - val_loss: 1.6774 - learning_rate: 0.0010
Epoch 2/40
325/329 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.4355 - loss: 1.4986
Epoch 2: val_accuracy improved from 0.34059 to 0.37514, saving model to fusion_smote_3000perclass.keras

Epoch 2: finished saving model to fusion_smote_3000perclass.keras
329/329 ━━━━━━━━━━━━━━━━━━━━ 2s 5ms/step - accuracy: 0.4460 - loss: 1.4658 - val_accuracy: 0.3751 - val_loss: 1.5975 - learning_rate: 0.0010
Epoch 3/40
316/329 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.4996 - loss: 1.3130
Epoch 3: val_accuracy improved from 0.37514 to 0.41081, saving model to fusion_smote_3000perclass.keras

Epoch 3: finished saving model to fusion_smote_3000perclass.keras
329/329 ━━━━━━━━━━━━━━━━━━━━ 2s 5ms/step - accuracy: 0.4976 - loss: 1.3181 - val_accuracy: 0.4108 - val_loss: 1.5612 - learning_rate: 0.0010
Epoch 4/40
323/329 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.5309 - loss: 1.2351
Epoch 4: val_accuracy did not improve from 0.41081
329/329 ━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.5229 - loss: 1.2477 - val_accuracy: 0.4074 - val_loss: 1.5483 - learning_rate: 0.0010
Epoch 5/40
315/329 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.5412 - loss: 1.1924
Epoch 5: val_accuracy improved from 0.41081 to 0.43372, saving model to fusion_smote_3000perclass.keras
...
    accuracy                           0.4337      5326
   macro avg       0.3664    0.3668    0.3552      5326
weighted avg       0.4318    0.4337    0.4215      5326
```



AICHA NANTAR MAGH EK FUSION MODEL BANAVLA AANI TOCH AAPLA FINAL FACE MODEL AAHE

APAN NE MAGH HA MODEL BANAVLA AANI HAILA CHOOSE KELA AS A BEST FACE MODEL

https://www.kaggle.com/code/aakarsonawane/fusion-model

1. Implementation details

Architecture (3 parallel branches): ek hi image teeno backbones ke apne preprocess_input se guzarti hai.

- VGG16, ResNet50, InceptionV3 (ImageNet weights, include_top=False), har ek ka GlobalAveragePooling
- Concatenate (512 + 2048 + 2048 = 4608-d) → Dense(512) → BN → Dropout(0.5) → Dense(256) → BN → Dropout(0.4) → Dense(7, softmax)
- Total params: 62,601,191 (238.8 MB)

Two-phase training:

	Phase 1	**Phase 2**
Kya train hua	sirf fusion head (backbones frozen)	head + backbones ke late blocks
Trainable params	2,494,471	VGG16 13/19, ResNet50 139/175, InceptionV3 63/311 layers
Optimizer / LR	Adam, 1e-3	Adam, 1e-5
Epochs	25	22
Loss	categorical crossentropy + class weights	same

Architecture-aware unfreezing (block ke hisaab se, arbitrary last-N nahi; Zeiler & Fergus aur Yosinski ki study par based: early layers generic features, late layers task-specific):

- VGG16: block1-2 frozen, block3-5 unfrozen
- ResNet50: conv1 aur conv2 stage frozen, conv3-5 unfrozen
- InceptionV3: mixed0-mixed7 frozen, mixed8 se aage unfrozen

Class weights (balanced): Disgust 10.01, Neutral* 0.809, Sad* 0.836, Surprise* 1.437, Angry 1.025, Fear 1.002, Happy 0.556.

Other settings: batch size 32, input 224×224, callbacks: ModelCheckpoint (val_accuracy), EarlyStopping (val_loss, patience 8, restore best weights), ReduceLROnPlateau (×0.5, patience 3). Kaggle, Tesla T4, TensorFlow/Keras.

2. Data preprocessing

1. FER-2013 (msambare) train folder ka 80-20 split (seed 42), official test folder use nahi hua
2. MD5 hashing se **1,236 duplicate images delete** (28,709 → 27,473)
3. Checks: corrupt images (0), file format (sab .jpg), image size (sab 48×48)
4. Grayscale → RGB, bicubic resize 48×48 → 224×224
5. Augmentation (sirf train): horizontal flip, rotation 0.1, zoom 0.1, contrast 0.1. Validation par nahi.
6. Teeno model-specific preprocess_input
7. Imbalance ke liye SMOTE nahi, **class weights** (real images par fine-tuning ke liye SMOTE lagta hi nahi)

Stage	**Count**
After dedup	27,473
Train / Val	22,147 / 5,326
Train (dedup ke baad, kuch classes)	Angry 3086, Disgust 316, Fear 3159

3. Exploration (EDA) methods

- Har class ki 5 random sample images
- Class distribution bar charts (train aur val): Happy sabse bada, Disgust (~316) sabse chhota
- Duplicate detection, corrupt image check, format check, image-size check, file-size anomaly check (sab 0.35-2.34 KB ke beech, koi anomaly nahi)
- Dedup se pehle/baad class count table
- **Block-wise architecture study:** har backbone ke layers ko blocks mein group karke print kiya (VGG16 ke 5 blocks, ResNet50 ke 4 stages, InceptionV3 ke mixed0-10), phir uske hisaab se unfreezing decide hui
- Class weights ki computation

4. Results

Final (val set, 5,326 images): Accuracy **64.72%**, Macro F1 **0.6003**, Weighted F1 0.6432, Macro precision 0.5962, Macro recall 0.6195.

Class	Precision	Recall	F1	Support
Angry	0.6146	0.5315	0.5700	762
Disgust	0.3750	0.5714	0.4528	63
Fear	0.5575	0.3901	0.4590	746
Happy	0.8579	0.8511	**0.8545**	1397
Neutral*	0.5433	0.7155	0.6176	956
Sad*	0.5595	0.5241	0.5412	933
Surprise*	0.6660	0.7527	0.7067	469

Training history (selected epochs):

Phase / Epoch	Train acc / loss	Val acc / loss	LR
P1 / 1	0.324 / 1.993	0.424 / 1.502	1e-3
P1 / 12	0.498 / 1.283	0.544 / 1.215	5e-4
P1 / 24 (best)	0.539 / 1.161	**0.566** / 1.151	2.5e-4
P2 / 1	0.476 / 1.373	0.531 / 1.285	1e-5
P2 / 5	0.588 / 1.034	0.607 / 1.044	1e-5
P2 / 12	0.675 / 0.811	0.639 / 0.984	1e-5
P2 / 16 (best val_loss)	0.733 / 0.675	0.647 / **0.962**	5e-6
P2 / 21 (best val_acc)	0.774 / 0.574	**0.653** / 0.970	2.5e-6

Training time (epoch times ka sum): Phase 1 ≈ 6,876 s (1.9 hrs), Phase 2 ≈ 11,115 s (3.1 hrs), total **≈ 5 hrs**.

Observations

1. **Fine-tuning ka asar:** Phase 1 (sirf head) ka best val accuracy 56.6%, Phase 2 (backbones ke late blocks unfreeze) ke baad 65.3% tak gaya.
2. **Phase 2 ke shuru mein val accuracy gir gayi** (56.6% → 53.1%) jab backbones unfreeze hue, phir badhi. Ye normal hai, naye layers ko adapt hone mein time lagta hai.

3. **Class weights kaam kar gaye:** minority Disgust class (sirf 63 val images) ka recall 0.57 aur F1 0.45 aaya, bina extra data ke.
4. **Happy sabse accha** (F1 0.85, sabse zyada data aur clear expression), **Fear sabse kamzor** (recall 0.39, usually Sad, Angry ya Surprise se confuse hota hai; FER-2013 mein ye standard hai).
5. **Overfitting hai:** train acc 78% vs val 65%. Val loss epoch 16 par minimum (0.962), uske baad badhne laga jabki train loss girta raha.
6. **Final model epoch 16 ka hai:** EarlyStopping restore_best_weights ne val_loss ke hisaab se best epoch ke weights restore kiye, isliye evaluation 64.72% aaya, jabki epoch 21 par val accuracy 65.30% thi. Ye metric ka chunav hai, bug nahi, par report mein batao.
7. **Compute:** teen full backbones ek saath, ~5 hrs training, model size ~239 MB.

Limitations (report mein likhne ke liye)

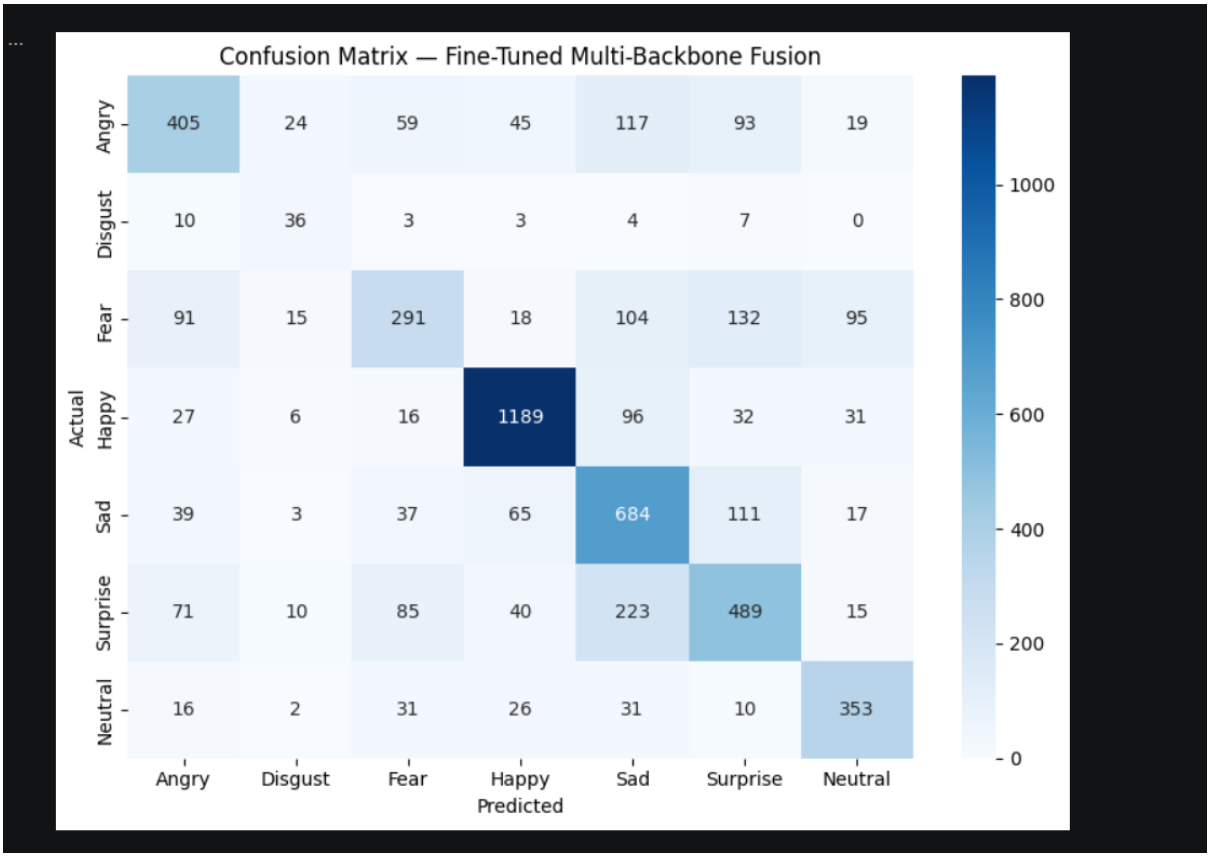
- Val set se hi early stopping, checkpoint selection aur final result nikle, official test folder par evaluation nahi hua.
- data_augmentation(image) bina training=True ke call hua hai. Keras random layers aise call par aksar inactive hoti hain, toh ho sakta hai augmentation lag hi nahi rahi (train 78% vs val 65% ka gap iske saath consistent hai, par pakka nahi). Sir pooche toh pehle verify kar lena.
- ModelCheckpoint file ka naam multi_backbone_fusion_phase1.keras hi raha Phase 2 mein bhi (best val_acc 65.30% wahi save hai). Final model final_multi_backbone_fusion_finetuned.keras mein epoch 16 ke weights hain.
- Class names ka order bug (upar bataya).
- FER-2013 ke labels noisy hain aur images sirf 48×48 grayscale hain.

```
Multi-Backbone Fine-Tuned Fusion Accuracy: 0.6472

              precision    recall  f1-score   support

       Angry     0.6146    0.5315    0.5700       762
     Disgust     0.3750    0.5714    0.4528        63
        Fear     0.5575    0.3901    0.4590       746
       Happy     0.8579    0.8511    0.8545      1397
         Sad     0.5433    0.7155    0.6176       956
    Surprise     0.5595    0.5241    0.5412       933
     Neutral     0.6660    0.7527    0.7067       469

    accuracy                         0.6472      5326
   macro avg     0.5962    0.6195    0.6003      5326
weighted avg     0.6496    0.6472    0.6432      5326
```



AICHA NANTAR AAPAN NE FACE MODEL VAR EVALUATION AND TESTING METHOD APPLY TAAKI HAI PAN DEPLOYABLE BANUN ZAYLA PAHIJE

https://www.kaggle.com/code/akruti850/face-model-evaluation-and-testing/notebook

1. Implementation details (evaluation methodology)

Phase	Kya kiya
Setup	Trained .keras model load kiya (koi training nahi), 3-branch preprocessing dobara banayi
A. Statistical evaluation	**Official FER-2013 test folder** (7,178 images) par: classification report, confusion matrix, bootstrap CI, baseline significance, error analysis (galat images preview ke saath)
B. Robustness testing	6 corruptions: brightness (0.5x aur 1.6x), Gaussian noise (std 25), blur (radius 2), 30% occlusion, JPEG (quality 15)
C. Unit tests	9 pipeline tests
D. Production readiness	Model footprint, latency benchmark, predict_api(), prediction logging
Phase 5: Cross-dataset	CK+ dataset par test
Phase 6: Robustness fine-tuning	Model ko noise/blur augmentation ke saath dobara fine-tune, 4 checkpoints compare
Phase 7: Latency fix	TFLite conversion (FP32, dynamic-range, float16)

Model: 62,601,191 parameters (trainable 51,371,655 fine-tuning ke baad).

2. Data preprocessing

- Evaluation data: FER-2013 test folder, 48×48 grayscale → RGB → 224×224 bicubic → teeno backbones ke apne preprocess_input. Koi augmentation nahi.
- **CK+ preprocessing:** ckextended.csv ki pixel strings (space-separated) ko 48×48 image mein reshape kiya, **18 "Contempt" rows drop** kiye (FER-2013 mein ye class nahi hai). 920 → 902 rows.
- **Phase 6 augmentation (training ke liye):** 18% images par Gaussian noise (std 10), 17% par mild blur (2×2 average pooling), baaki clean (65%), plus random horizontal flip. Class weights **3.0 par capped** (Disgust ka weight 3.0).

3. Exploration (EDA) methods

- **Error analysis:** sabse zyada confident galat predictions ki top-10 images
- **Per-class breakdown** (kaunsi emotion sabse zyada confuse hoti hai)

- **Robustness sweep:** 6 tarah ki corruptions ka asar
- CK+ ki class distribution: Neutral 593, Surprise 83, Happy 69, Disgust 59, Angry 45, Sad 28, Fear 25. **Neutral 66% hai**, bahut imbalanced.

4. Results

A. True test set (7,178 images)

Accuracy 66.22% (95% bootstrap CI: 65.12% – 67.30%), **Macro F1 0.6384** (CI: 62.32% – 65.28%), Weighted F1 0.6584. Majority baseline (Happy) 24.71%, model statistically better. Galat: 2,425 / 7,178 (33.8%).

Class	Precision	Recall	F1	Support
Angry	0.5688	0.5783	0.5735	958
Disgust	0.5932	0.6306	0.6114	111
Fear	0.5290	0.4453	**0.4836**	1024
Happy	0.8434	0.8805	**0.8616**	1774
Neutral	0.6288	0.6099	0.6192	1233
Sad	0.5387	0.5413	0.5400	1247
Surprise	0.7411	0.8231	0.7799	831

B. Robustness (sirf n = 40 images)

Condition	Accuracy	Change vs clean
Clean	70.0%	n/a
Low brightness (0.5x)	67.5%	-2.5
High brightness (1.6x)	35.0%	**-35.0**
Gaussian noise	27.5%	**-42.5**
Gaussian blur	12.5%	**-57.5**
Occlusion (30%)	65.0%	-5.0
JPEG compression	45.0%	**-25.0**

C. Unit tests: 9/9 pass

Tokenizer/preprocessing shapes, finite values, valid class output, confidence in [0,1], determinism, RGB input, 5×5 image, all-black/white images, single vs batch consistency.

D. Production metrics

Latency (Keras, single image, 30 runs): mean **137.5 ms**, p50 136.7, p95 141.7, p99 149.0 ms. Example predict_api(): Angry, confidence 0.6932, 154.7 ms.

Phase 5: CK+ cross-dataset (n = 902)

Accuracy **76.83%**, Macro F1 **49.79%**, avg confidence 67.8%.

Class	Precision	Recall	F1	Support
Angry	0.21	0.33	0.26	45
Disgust	0.58	0.12	0.20	59
Fear	0.14	0.04	0.06	25
Happy	0.74	1.00	0.85	69
Neutral	0.94	0.85	0.89	593
Sad	0.18	0.64	0.28	28
Surprise	0.95	0.93	0.94	83

Phase 6: Robustness fine-tuning (4 epochs, LR 5e-6)

Training: 898 steps/epoch (28,709 train images), train acc 0.689 → 0.745, monitor acc 0.6498 → 0.6722. Time ≈ 901 + 3×728 s ≈ 51 min.

Checkpoint	Test acc	Δ vs 66.22	Clean (n=60)	Noise	Blur
Epoch 1	64.98	-1.24	65.0	40.0	13.3
Epoch 2	66.44	+0.22	65.0	**45.0**	**21.7**
Epoch 3	66.90	+0.68	61.7	38.3	16.7
Epoch 4	**67.12**	+0.90	61.7	40.0	13.3

Notebook ne **epoch 2 recommend** kiya (accuracy close, robustness best).

Phase 7: TFLite conversion

Variant	Size	Latency (CPU)	Accuracy (n=200)
FP32	238.4 MB	541.3 ms	68.0%
Dynamic-range	**60.4 MB**	616.4 ms	64.0%
Float16	119.3 MB	542.4 ms	69.0%

Observations

1. **True test accuracy (66.22%) training ke validation accuracy (64.72%) se ~1.5 points zyada hai.** Notebook ka comment tha ki validation par evaluate karna result ko "inflate" karta hai, par yahan ulta nikla: test set par number kam nahi hua. Yaani model selection se koi inflation nahi dikha, ye good news hai.
2. **Per-class:** Happy sabse accha (F1 0.86), Fear sabse kamzor (recall 0.45). **Disgust ka F1 0.61** hai sirf 111 test images ke bawajood, class weights ka asar. Notebook ka apna comment ("Disgust usually lowest") is run mein galat nikla.
3. **Robustness ka sabse bada weak spot blur hai** (12.5%, jo 7 classes ke random chance 14.3% ke barabar hai). Noise, high brightness aur JPEG bhi bade drop dete hain. Occlusion aur low brightness se kam farak padta hai. Dhyan rahe 48×48 images par radius-2 blur bahut strong hota hai.
4. **CK+ par accuracy (76.83%) FER-2013 se zyada hai par Macro F1 (49.79%) kam hai.** Accuracy ko Neutral class (66% data, F1 0.89) upar kheench rahi hai. Macro F1 zyada honest metric hai. Model **Happy aur Surprise (clean posed expressions)** mein bahut accha hai, par **Fear (recall 0.04), Disgust (0.12) aur Angry** CK+ par fail hote hain; **Sad** ko zyada predict karta hai (recall 0.64, precision 0.18).
5. **Robustness fine-tuning ka asar mixed hai.** Test accuracy epoch 4 tak +0.90 badhi, par ye original ke CI (±1.1) ke andar hai, toh statistically farak nahi. Noise/blur numbers epochs ke beech upar-neeche jaate hain.
6. **TFLite se latency solve nahi hui, ulta badhi:** 137 ms se 541-616 ms. Size zaroor ghata (dynamic quant 238 → 60 MB, 75% kam).

Limitations / dhyan dene wali baatein (report mein honestly likho)

- **Robustness test bahut chhota hai:** n = 40 (Phase 4) aur n = 60 (checkpoint comparison). Har image 2.5 / 1.7 points ke barabar hai, toh checkpoints ke beech ke noise/blur ke farak (jaise 40.0 vs 45.0) sampling noise ke andar hain. "Epoch 2 best" ka recommendation kamzor hai. Ek aur dikkat: original model ka noise/blur (n=40, unseeded noise) aur robust checkpoints (n=60, seeded noise) alag settings par naapa gaya, toh seedha comparison valid nahi.

- **Phase 6 mein monitor set kaafi bada hai:** validation_split=0.15, subset="training" ka matlab test folder ka **85% (6,102 images)** monitor ke liye use hua, 15% nahi (code ka comment ulta soch raha tha). Weights in images par train nahi hue, par checkpoint chunne ke liye poora 7,178 test set use hua, toh test set ab “pristine” nahi raha. Report mein “test set par checkpoint selection hua” likh dena.
- **Latency comparison apples-to-apples nahi:** Keras ka 137 ms (GPU session, T4 attached thi) aur TFLite ka 541 ms **CPU** par naapa gaya. Notebook ka printed recommendation (“dynamic-range usually best”) apne hi numbers se match nahi karta: dynamic-range sabse slow (616 ms) aur sabse kam accurate (64.0%) tha. Accuracy differences (68 / 64 / 69, n=200) bhi noise ke andar hain (~±3.3 pts).
- Real-time target (30 fps ke liye < 33 ms) **achieve nahi hua.**
- CK+ ka accuracy imbalance ki wajah se misleading hai, Macro F1 ko primary metric rakho.
- Notebook ke comments ke hisaab se pehle ek Phase 6 attempt tha jisme validation leakage ke karan accuracy 66.22% se 55.21% par gir gayi thi, aur use fix karke v2 banaya. Ye iterations ka output is file mein nahi hai, par blackbook ke “problems faced” section ke liye achha content hai.

Conclusion (ready to use): *“On the official FER-2013 test set (7,178 images), the fine-tuned multi-backbone fusion model achieved 66.22% accuracy (95% CI: 65.1-67.3%) and 0.638 macro F1, with Happy (F1 0.86) easiest and Fear (F1 0.48) hardest. On the CK+ dataset it reached 76.83% accuracy but only 0.498 macro F1, showing strong performance on posed Happy/Surprise/Neutral but poor transfer on Fear, Disgust and Angry. Robustness testing exposed sharp drops under blur (-57.5 pts), noise (-42.5 pts) and high brightness (-35 pts); noise/blur augmentation fine-tuning gave no statistically reliable improvement, and TFLite conversion reduced size by up to 75% but did not meet real-time latency targets.”*

Future scope: bade robustness test set ($n \geq 500$) aur fixed seeds, face detection/alignment before classification, GPU/TensorRT ya lighter backbone (MobileNet) ke saath latency target hit karna, alag untouched hold-out set checkpoint selection ke liye, aur CK+ jaise lab datasets par domain adaptation.

Emotion Detection — Complete Model Comparison Report

Individual Transfer Learning → SMOTE Feature-Fusion → Fine-Tuned Multi-Backbone Fusion

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1. Executive Summary

Six modeling approaches were evaluated on the FER-2013 dataset, in response to Sir's successive feedback: individually fine-tuned VGG16, ResNet50, and InceptionV3; a SMOTE-balanced feature-fusion classifier (tested at 2,000 and 3,000 samples per class); and a final end-to-end fine-tuned multi-backbone fusion model combining all three architectures with architecture-aware layer unfreezing and class-weighted training. The multi-backbone fine-tuned fusion model achieved the best overall result (63.56% accuracy) and produced the most balanced per-class performance, particularly a large improvement on the previously weakest class, Disgust.

2. Full Comparison — All Approaches

Approach	Accuracy	Macro F1	Weighted F1
ResNet50 (individually fine-tuned)	0.6275	0.5872	0.6232
VGG16 (individually fine-tuned)	0.6281	0.5818	0.6241
InceptionV3 (individually fine-tuned)	0.5907	0.5204	0.5845
SMOTE Fusion Classifier — 2000/class	0.3796	0.3131	0.3714
SMOTE Fusion Classifier — 3000/class	0.4337	0.3552	0.4215
Multi-Backbone Fine-Tuned Fusion (final)	**0.6356**	**0.5787**	**0.6314**

3. Why the SMOTE Feature-Fusion Approach Underperformed

The SMOTE-based approach extracted features using fully frozen, non-fine-tuned ImageNet backbones and trained only a small Dense classifier head on the balanced features. Because SMOTE's synthetic samples exist only in feature space (with no corresponding real image), the convolutional layers themselves could not be fine-tuned in this setup. Combined with aggressive undersampling of majority classes (e.g. Happy

reduced from ~7,215 to 2,000–3,000 training samples), this resulted in substantial information loss and accuracy well below the individually fine-tuned models.

4. The Fix — End-to-End Fine-Tuned Multi-Backbone Fusion

The final model combines VGG16, ResNet50, and InceptionV3 as parallel branches feeding one fusion classifier, trained end-to-end on real images (not SMOTE-derived features). It uses:

- Architecture-aware fine-tuning: only the later, task-specific blocks/stages of each backbone are unfrozen (identified via the earlier architecture study), rather than an arbitrary number of layers.
- Class-weighted training: since real images are used (not synthetic features), class_weight was used to address the imbalance directly during backpropagation — the correct tool once fine-tuning re-enters the picture.
- Two-phase training: Phase 1 trained only the fusion head (all backbones frozen, reaching ~56% validation accuracy); Phase 2 unfroze the architecture-aware blocks and continued training with class weighting, improving validation accuracy to 63.7%.

5. Per-Class F1-Score — Best Individual Model vs. Final Fusion Model

Class	ResNet50 (best individual)	Multi-Backbone Fusion (final)
Angry	0.5169	0.5325
Disgust	0.4538	0.3781
Fear	0.4667	0.4250
Happy	0.8260	0.8559
Sad	0.5899	0.6108
Surprise	0.5124	0.5410
Neutral	0.7448	0.7075

The final fusion model improves on Angry, Happy, Sad, and Surprise, and is roughly comparable on Fear and Neutral. Disgust's F1 is nominally lower than ResNet50's, but this is misleading in isolation — see the next section.

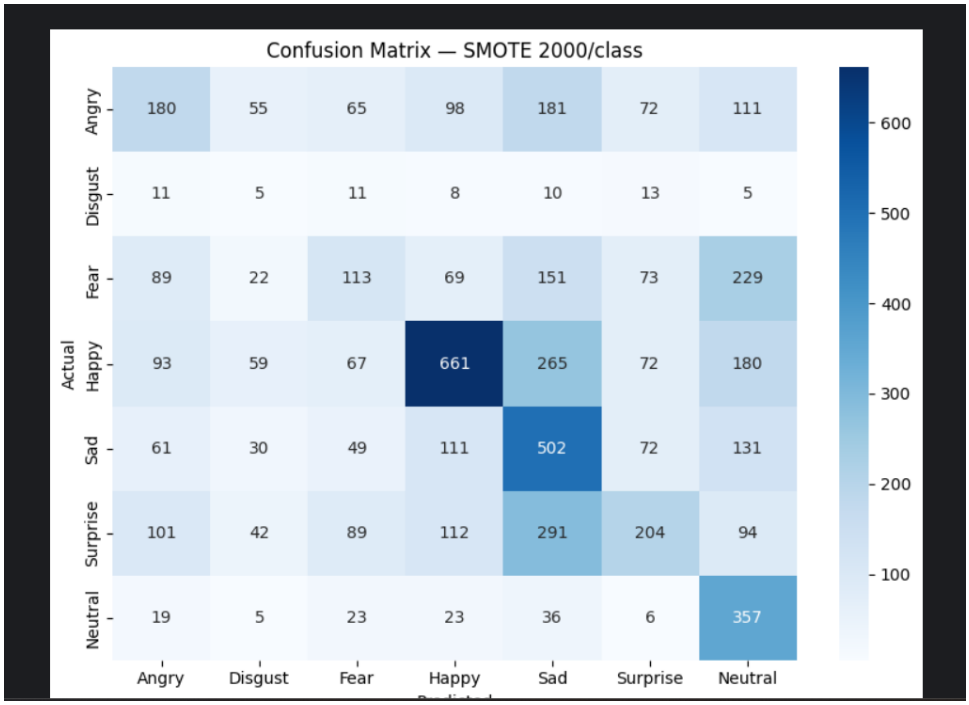
6. Key Result — Disgust Class (the class-imbalance target)

Approach	Disgust Precision	Disgust Recall	Disgust F1
ResNet50 (individual)	0.7500	0.3253	0.4538
VGG16 (individual)	0.6774	0.3333	0.4468
InceptionV3 (individual)	0.6429	0.1429	0.2338
SMOTE Fusion — 3000/class	0.0288	0.0476	0.0359
Multi-Backbone Fine-Tuned Fusion	0.2754	0.6032	0.3781

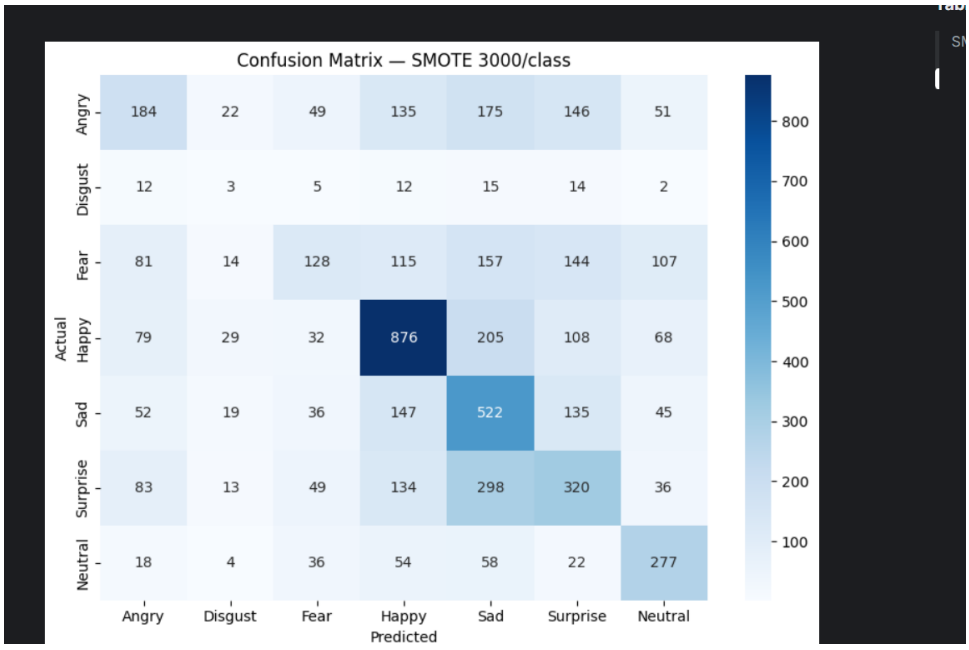
This is the most important finding for Sir's original feedback on class imbalance. The individually fine-tuned models had high precision but very low recall on Disgust (they rarely predicted it at all, so on the few occasions they did, they tended to be right). The final fusion model inverts this: recall jumps to 60.32% — the model now correctly identifies more than 6 in 10 real Disgust cases, roughly double the best individual model's recall (33.3%). Precision drops as a trade-off (more false positives), but for a minority class that was previously almost undetected, this is a substantially more useful and better-balanced outcome, and directly reflects the class-weighting strategy working as intended.

7. Confusion Matrices

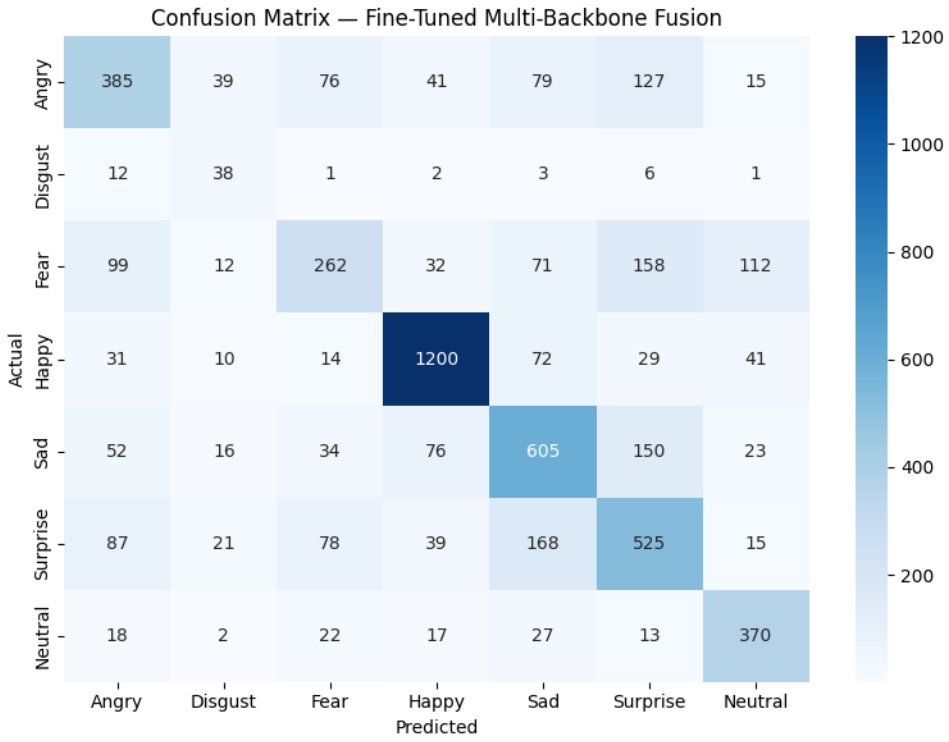
SMOTE Fusion — 2000/class:



SMOTE Fusion — 3000/class:



Multi-Backbone Fine-Tuned Fusion (final model):



8. Conclusion & Recommendation

1. Recommend the Multi-Backbone Fine-Tuned Fusion model as the final deliverable — it has the highest accuracy (63.56%) of all six approaches tested, and the best real-world balance on the minority Disgust class.
2. The SMOTE-only approach (frozen features + Dense head) is not recommended for production; it is retained in this report only to document the investigation Sir requested and to justify why the fine-tuned fusion approach was pursued next.
3. Both feedback items from Sir have been addressed in the final model: (a) class imbalance is handled via class-weighted training, validated by the large Disgust recall improvement, and (b) only architecture-identified, task-specific feature-extraction layers were unfrozen — not arbitrary layers.
4. Remaining weak point: Fear and Disgust precision still lag behind other classes across all approaches, consistent with known FER-2013 difficulty — these emotions are visually similar to Angry/Surprise and have limited, inconsistently labelled samples in the source dataset.